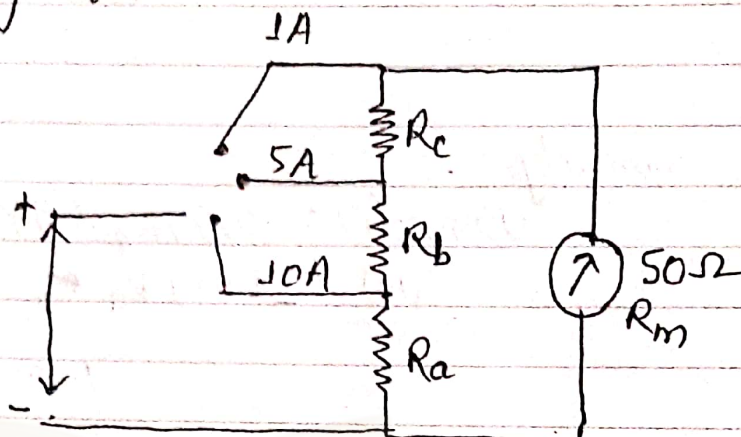


- ④ Design an Ayton shunt to provide an ammeter with current ranges of 1A, 5A and 10A. A D'Arsonval movement with an internal resistance $R_m = 50\Omega$ and full scale deflection current of 1mA is used in the configuration.



⇒ On the 1A range:

$R_a + R_b + R_c$ are in parallel with 50Ω movement. Since the coil requires 1mA for full scale deflection, the shunt will be required to pass a current of $1A - 1mA = 999mA$.

So,

$$(R_a + R_b + R_c) \times 999 = 50 \times 1$$

$$\therefore R_a + R_b + R_c = \frac{50}{999} = 0.05005\Omega \quad \text{--- (i)}$$

On the 5A range:

$R_a + R_b$ are in parallel with $R_c + 50\Omega$. In this case 1mA current flows thru $(R_c + R_m)$ and 4999mA flows thru $R_a + R_b$. So,

$$(R_a + R_b) 4999 = (R_c + 50) 1$$

$$\therefore R_a + R_b = \frac{R_c + 50}{4999} \quad \text{--- (ii)}$$

On the 10A range,

R_a is in parallel with $R_b + R_c + R_m$. In this case 1mA flows thru $(R_b + R_c + R_m)$ & 9999mA flows thru R_a . So

$$R_a \times 9999 = (R_b + R_c + R_m) \times 1$$

$$\therefore R_a = \frac{R_b + R_c + R_m}{9999} \quad \text{--- (iii)}$$

Solving these three simultaneous eqn (i), (ii) & (iii), we get,

$$4999 x(i) : 4999 R_a + 4999 R_b + 4999 R_c = 250.2$$

$$(ii) : \underline{4999 R_a + 4999 R_b - R_c = 50}$$

$$5000 R_c = 200$$

$$\therefore R_c = 0.04004 \Omega$$

Similarly,

$$9999 x(i) : 9999 R_a + 9999 R_b + 9999 R_c = 500.45$$

$$(iii) : \underline{9999 R_a + R_b + R_c = 50}$$

$$10,000 R_b + 10,000 R_c = 450.45$$

$$\text{or, } 10,000 R_b + 10,000 \times 0.04004 = 450.45$$

$$\therefore R_b = 0.005005 \Omega$$

$$\& R_a = 0.005005$$

The calculation indicates that for larger currents, the value of shunt resistors become very small.